LLM-Guided Grammar-Aware Fuzzing of PostgreSQL

Progress meeting 2: what we executed, what we found

CS-5187 term project A3 · Venu · Bhuvan · Gowtham · 29 October 2026

RESULT 1

Upstream bug scorecard

Member	Bugs filed	Replied: confirmed a bug	Fixed upstream
Venu	2	0	0
Gowtham	3	**3**	**2**
Bhuvan	4	1	0
Team total	**9**	**4**	**2**

9

bugs filed on pgsql-bugs / pgsql-hackers

4

answered by committers or contributors

2

already fixed on PostgreSQL master

Bugs were found by our fuzzer, by oracle-guided audits of recent PostgreSQL code, and by reducing campaign findings.

BUG DETAILS

Venu: 2 filed

Bug	Upstream	What it is (short note)	Status
BUG-01 JSON_TABLE collations	#19768	JSON_TABLE never assigns collations to its context item and PASSING args. Assertion in execExpr.c on master, silent wrong result on 18.6. Found by the grammar fuzzer; one-line patch + tests.	**filed 16 Oct; no reply yet**
BUG-02 range containment with NULL	#19793	Planner support function folds range @> NULL-element to a constant, so a strict operator returns true/false instead of NULL. Silent wrong result in released 17 and 18. Found by our (SELECT e) rewrite oracle.	**filed 23 Oct; patch sent; no reply yet**

BUG DETAILS

Gowtham: 3 filed, 3 answered, 2 fixed

Bug	Upstream	What it is (short note)	Status
MERGE: SubPlan with no parent plan	#19802	MERGE into a partitioned table with a non-pulled-up source and a scalar subquery in UPDATE SET fails with an internal error. Found by the grammar arm.	**confirmed by a committer; fix posted**
Parallel Append assertion	#19803	Assertion failure in a parallel Append plan, found through a TLP check by the LLM-fragment arm.	**confirmed; fixed on master (abda2a2c)**
to_char NaN / NORMALIZE crash	#19771	to_char on NaN leads to an assertion in NORMALIZE. Reduced by our tool to 27 tokens.	**fix committed (776df8a6), authored by Gowtham**

BUG DETAILS

Bhuvan: 4 filed, 1 answered

Bug	Upstream	What it is (short note)	Status
AUDIT-F1 NOT IN anti-join cast	pgsql-bugs	Wrong-result regression from commit 383eb21ebf: NOT IN is turned into an anti-join and ignores an implicit cast on the sub-select side. In 19rc1 and master, not in 18.6. Patch + test.	**filed 23 Oct; awaiting reply**
AUDIT-F2 IGNORE NULLS hang	pgsql-bugs	Frame scan of window functions with IGNORE NULLS can hang.	**filed 23 Oct; Ishii replied 24 Oct**
C-01 dependencies.c assertion	pgsql-bugs	Duplicate expression in extended statistics trips an assertion. Reproducer + patch.	**filed 27 Oct; awaiting number**
Extended stats non-finite MCV	pgsql-hackers	pg_restore_extended_stats accepts non-finite MCV frequencies. Patch submitted.	**patch under review**

RESULT 1

Evidence: BUG-02 reproduced on 18.6 and master

```
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ PSQL=~/pgfuzz/install/REL_18_STABLE/bin/psql
for p in 5435 5440; do
  $PSQL -h 127.0.0.1 -p $p -U postgres \
    -c "SELECT version()" \
    -c "SELECT c, '(,)'::int4range @> c AS a, 'empty'::int4range @> c AS b, NOT ('empty'::int4range @> c) AS d FROM (VALUES (NULL::int), (1)) v(c);" \
    -c "EXPLAIN VERBOSE SELECT '(,)'::int4range @> c FROM (VALUES (NULL::int), (1)) v(c);"
done
                                                version

--------------------------------------------------------------------------------
--------------------------
 PostgreSQL 18.6 on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubunt
u2~24.04.1) 13.3.0, 64-bit
(1 row)

 c | a | b | d
---+---+---+---
   | t | f | t
 1 | t | f | t
(2 rows)

                        QUERY PLAN
-----------------------------------------------------------
 Values Scan on "*VALUES*"  (cost=0.00..0.03 rows=2 width=1)
   Output: true
(2 rows)

                                                version

--------------------------------------------------------------------------------
----------------------------
 PostgreSQL 20devel on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ub
untu2~24.04.1) 13.3.0, 64-bit
(1 row)

 c | a | b | d
---+---+---+---
   | t | f | t
 1 | t | f | t
(2 rows)

                        QUERY PLAN
-----------------------------------------------------------
 Values Scan on "*VALUES*"  (cost=0.00..0.03 rows=2 width=1)
   Output: true
(2 rows)
```

What you see

- NULL row: every column must be NULL
- 18.6 and master print t / f / t
- EXPLAIN shows Output: true, the planner folded it

How we found it

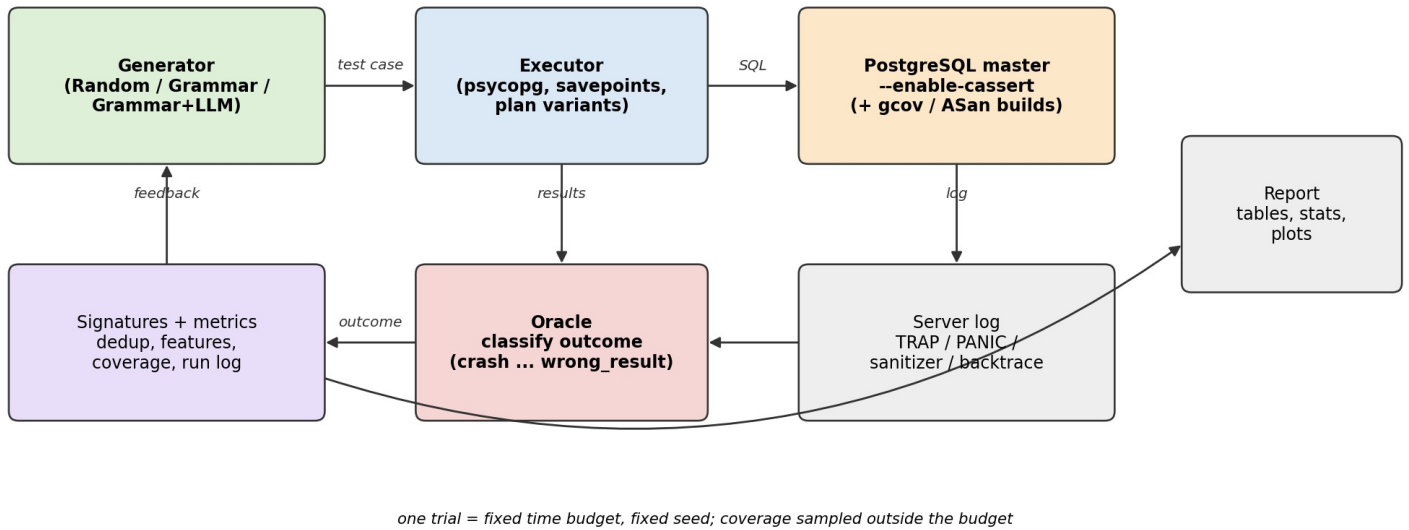
- Oracle rewrites e as (SELECT e) on the derivation tree
- Found within the first 150 test cases the rewrite ran
- Root cause: rangetypes.c support function; patch sent for 17, 18, master

EXECUTION

Everything planned at meeting 1 is built and run

Plan item	What we executed	Status
Typed grammar	78 rules, 421 alternatives, 13 types; derivation trees with node spans	done
Metamorphic oracles	12 kinds: TLP (WHERE, HAVING, aggregates), NoREC, expression/predicate rewrites, RLS leak, CERT	done
Plan + version oracles	18 plan variants; master vs 18.6 cross-version	done
Triage	reproduce x3, tree + token reduction, localization, upstream report draft	done
LLM extension	commit miner, prompt, validation gate, repair rounds, rewrite gate	done
Benchmark (RQ3)	23 reverted real fixes; fixed / solo / injected servers	done
Campaigns	7 arms x 5 trials x 1 h on two machines (RQ1-RQ4)	done
Quality	148 unit tests; about 10,500 lines of Python	done

How it works: one tree, four jobs



The derivation tree

- Generation builds it
- Mutation re-derives one subtree
- Oracles rewrite one typed node
- Triage reduces it

Why it matters

Because every node has a type and a position, an oracle can rewrite a single node at any query level, so queries stay valid SQL and the rewrites stay sound.

Evidence: from failure to upstream report

```
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ cat ~/cs5187-term-project/runs/memberA-rq3/logs/A3_triage.txt
[cde9712bc8db] assertion via assertion: reproducing x3
  statements: 3 -> 2
  statements: 2 -> 1
  clauses: 572 -> 570 tokens
  clauses: 570 -> 543 tokens
  clauses: 543 -> 529 tokens
  clauses: 529 -> 527 tokens
  clauses: 527 -> 515 tokens
  items: 515 -> 420 tokens
  items: 420 -> 410 tokens
  items: 410 -> 318 tokens
  items: 318 -> 314 tokens
  items: 314 -> 290 tokens
  items: 290 -> 277 tokens
  items: 277 -> 273 tokens
  items: 273 -> 269 tokens
  items: 269 -> 265 tokens
  items: 265 -> 263 tokens
  items: 263 -> 261 tokens
  groups: 261 -> 201 tokens
  groups: 201 -> 162 tokens
  groups: 162 -> 141 tokens
  groups: 141 -> 139 tokens
  groups: 139 -> 125 tokens
  groups: 125 -> 116 tokens
  reduced to 1 statement(s), 116 tokens in the culprit (406 runs, 73.3 s)
1 candidate(s) triaged into ../results/triage_demo; verdict template: ../results/triage_demo/verdicts.template.yaml
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$
```

pgfuzz triage on a real candidate

```
bhuvi@BlazingBhuvi:/mnt/c/Users/Bhuva/OneDrive/Desktop/cs5187-term-project-main$ cat ~/cs5187-term-project/runs/memberA-rq3/triage_demo/cde9712bc8db/upstream_report
.txt 2>&1
Subject: fails an assertion -- <one line: the feature involved>

PostgreSQL version: 20devel (built with --enable-cassert)
Also checked on: <latest release: run the script there too>
Platform: <OS, compiler>

Description
-----------

The following script fails an assertion.

TRAP: failed Assert("list_length(rcexpr->inputcollids) == nopers"), File: "execExpr.c", Line: 2078

Script
------

-- pgfuzz triage of cde9712bc8db: assertion via assertion
SELECT CASE CASE WHEN NULL THEN NULL ELSE (SELECT MAX ('{NULL}'::text[]) FROM JSON_TABLE (CASE WHEN (ROW (10.0::numeric, 10) >= ROW (0.0::numeric, 1) IS FALSE) THEN
 '{"k": -1, "a": [1, 2, 3]}'::jsonb ELSE '{"k": 1, "a": [1, 2]}'::jsonb END, '$.a[*]' COLUMNS ( c1 int PATH '$')) AS a6) END WHEN NULL THEN NULL WHEN MAX ('{a,b,c,d
}'::text[]) OVER (PARTITION BY '{1,2,3}'::int[]) THEN NULL ELSE '{NULL,1}'::int[] END AS c0 FROM (SELECT a1.c0 AS c0 FROM t0 AS a1) AS a2;

Expected
--------

<what should happen, and why: SQL standard / documentation section>

<If bisected: first bad commit (scripts/bisect_pg.sh).>
bhuvi@BlazingBhuvi:/mnt/c/Users/Bhuva/OneDrive/Desktop/cs5187-term-project-main$
```

generated upstream report draft

590 → 79

tokens after tree + token reduction of a BUG-01 case (tokens alone: 116)

x3

every candidate is reproduced three times on a fresh connection

plan x rewrite

localization names the planner path or the always-applied rewrite

The campaigns we ran

7 arms

Random, Grammar, +Oracles, no-bandit, +LLM fragments, +LLM seeds, +LLM both

5 x 1 h

trials per arm, same seeds, 4 workers, assertions on

2 laptops

RQ1/RQ2/RQ4 on one, RQ3 on the other

26bd08fb

one PostgreSQL master commit for everything

```
gow@Gowtham:~/cs5187-term-project/runs/memberB-rq1-llm$ sed -n '/Median (IQR)/,/^$/p' report/report.md
## Median (IQR) per configuration

| Metric | Random/Mutation | Grammar | Grammar+Oracles | Grammar+Oracles, no bandit | +LLM fragments | +LLM seeds | Grammar+Oracles+LLM |
|---|---:|---:|---:|---:|---:|---:|---:|
| Line coverage | 22.2% (21.9%-22.2%) | 36.7% (35.9%-36.7%) | 36.2% (36.1%-36.4%) | 36.2% (35.9%-36.4%) | 35.7% (35.7%-35.8%) | 35.6% (35.4%-35.8%) | 35.7% (35.6%-35.8%) |
| Branch coverage | 15.1% (15.0%-15.1%) | 27.1% (26.6%-27.2%) | 26.7% (26.6%-27.0%) | 26.8% (26.8%-26.9%) | 26.5% (26.4%-26.5%) | 26.4% (26.3%-26.5%) | 26.4% (26.4%-26.5%) |
| SQL features | 23 (22-24) | 101 (101-101) | 101 (101-101) | 101 (101-101) | 104 (104-104) | 106 (106-106) | 106 (106-106) |
| SQL feature coverage | 15.5% (14.9%-16.2%) | 68.2% (68.2%-68.2%) | 68.2% (68.2%-68.2%) | 68.2% (68.2%-68.2%) | 68.2% (68.2%-68.2%) | 70.3% (70.3%-70.3%) | 71.6% (71.6%-71.6%) | 71.6% (71.6%-71.6%) |
| Feature pairs | 57 (57-58) | 2,517 (2,515-2,519) | 2,471 (2,462-2,472) | 2,517 (2,515-2,517) | 2,461 (2,450-2,471) | 2,484 (2,476-2,501) | 2,457 (2,439-2,473) |
| Distinct query shapes | 8,407 (7,701-8,885) | 80,477 (79,052-81,785) | 73,157 (57,767-77,330) | 75,216 (71,260-83,772) | 55,056 (49,914-58,934) | 75,506 (68,180-81,487) | 53,454 (50,402-72,506) |
| Distinct plan shapes | 43 (42-45) | 20,279 (20,049-21,354) | 34,313 (30,524-36,909) | 23,997 (21,187-24,869) | 25,972 (25,281-28,017) | 29,274 (28,875-36,256) | 26,375 (24,497-35,385) |
| Metamorphic checks run | - | - | 41,740 (36,348-45,612) | 31,491 (29,585-36,088) | 33,230 (28,932-34,763) | 42,488 (40,151-45,633) | 30,520 (30,273-41,184) |
| Plan-variant runs | 154,440 (134,264-173,100) | 105,372 (101,616-108,018) | 166,718 (145,956-182,410) | 125,192 (118,708-142,328) | 131,438 (115,686-137,914) | 171,456 (162,560-184,904) | 124,416 (122,648-165,884) |
| Parse-valid rate | 33.6% (33.6%-33.7%) | 100.0% (100.0%-100.0%) | 100.0% (100.0%-100.0%) | 100.0% (100.0%-100.0%) | 99.9% (99.8%-99.9%) | 100.0% (100.0%-100.0%) | 99.9% (99.9%-99.9%) |
| Valid-query rate | 26.0% (26.0%-26.0%) | 99.3% (82.6%-99.5%) | 96.3% (96.0%-96.6%) | 99.4% (99.3%-99.4%) | 95.5% (95.4%-96.1%) | 98.3% (98.3%-98.6%) | 95.3% (94.5%-95.7%) |
| Statements/s | 298.4 (260.6-335.8) | 67.2 (65.2-69.9) | 39.9 (38.3-43.8) | 45.0 (43.3-51.8) | 32.5 (29.3-33.6) | 42.8 (42.0-49.1) | 31.0 (30.3-43.3) |
| Test cases | 1,074,240 (938,252-1,209,074) | 71,228 (70,105-72,835) | 58,601 (53,593-66,738) | 64,923 (60,977-72,797) | 48,222 (41,687-50,046) | 63,655 (61,591-69,453) | 46,754 (45,779-62,292) |
| Unique bug candidates | 0 (0-0) | 1 (1-2) | 1 (1-3) | 1 (1-2) | 1 (0-2) | 3 (1-4) | 1 (0-2) |
| Known bugs reproduced | 0 (0-0) | 0 (0-0) | 0 (0-0) | 0 (0-0) | 0 (0-0) | 0 (0-0) | 0 (0-0) |
| Time to first bug (s) | - | 1,514.6 (1,090.7-2,027.7) | 1,956.4 (1,172.0-2,133.0) | 470.9 (211.7-2,184.9) | 280.0 (236.4-1,076.4) | 598.6 (373.7-740.9) | 953.9 (546.2-1,516.8) |
```

campaign summary produced by the report tool

Statistics: medians, Mann-Whitney U, Vargha-Delaney A12 (with 5 trials the smallest possible p is 0.008).

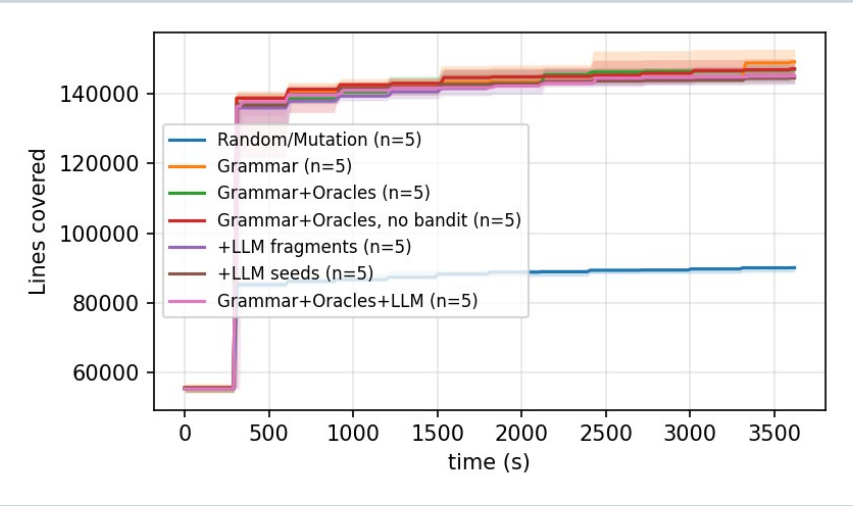
RESULTS

RQ1: the typed grammar reaches far more of PostgreSQL

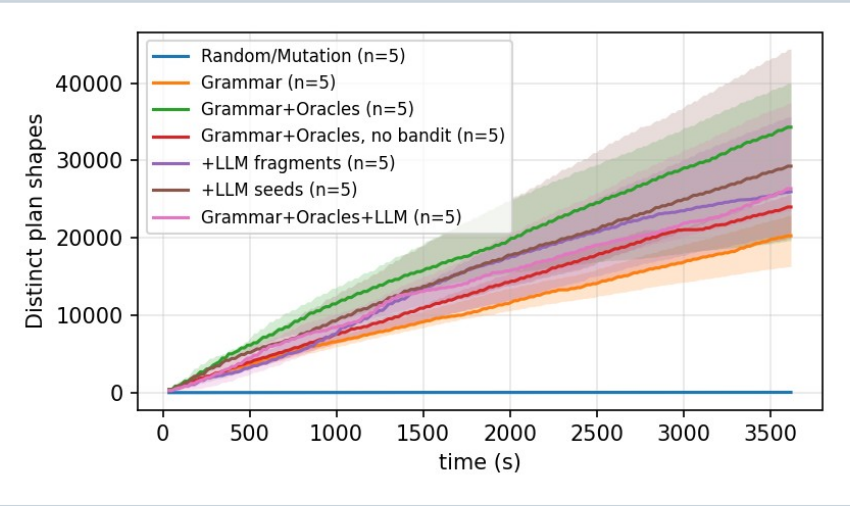
Arm	Valid	Lines	Features	Plan shapes	Tests
Random / mutation	**26.0%**	22.2%	23	43	1,074k
Grammar	**99.3%**	**36.7%**	101	20,279	71k
Grammar + oracles	96.3%	36.2%	101	**34,313**	59k
+ LLM (both)	95.3%	35.7%	106	26,375	47k

Reading

- Grammar vs random: A12 = 1.00, p = 0.008
- Oracles double plan shapes
- LLM adds features, not lines



backend line coverage over time



distinct plan shapes over time

Cost

Random runs 15x more tests and still reaches less.

RESULTS

RQ2: 93 candidates triaged, every one has a verdict

Verdict	Count
real bugs	7
duplicates of real upstream bugs	15
false positives	37
flaky (did not reproduce x3)	28
open, to verify	6

0.16

precision, duplicates not counted

0.37

precision if duplicates count

10 false-positive classes, all fixed with tests

- 6 found during development (ARRAY(SELECT), TLP-HAVING, numeric scale, seed clock, PoC replay, intended corr change)
- Equal intervals printed differently: 16 candidates
- ORDER BY NULL in ordered aggregates: 2
- Cross-version state divergence (IGNORE NULLS): 5
- LLM rewrite that changed the collation: 1
- Lesson: oracle precision, not generator power, is the bottleneck

RQ3: 23 real PostgreSQL bugs put back, and what each arm found

```
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ python3 -m pgfuzz bench verify configs/rq3.yaml
| bug | commit | symptom | all injected | evidence |
|---|---|---|---|---|
| rb-11b56ee2 | 11b56ee2 | assertion | assertion | pathkeys.c: var->varno == rel->relid |
| rb-94e1385e | 94e1385e | wrong_result | wrong_result | stmt 0: 1 rows on fixed, 1 on buggy |
| rb-41180127 | 41180127 | missing_error | missing_error | stmt 0: fixed raises 22030 duplicate JSON object key value |
| rb-b28a2b2b | b28a2b2b | wrong_result | wrong_result | stmt 5: 1 rows on fixed, 1 on buggy |
| rb-f5635c27 | f5635c27 | none | none |  |
| rb-d8444163 | d8444163 | wrong_result | wrong_result | stmt 3: 0 rows on fixed, 1 on buggy |
| rb-62e46c00 | 62e46c00 | error | error | stmt 2: 42P10 there is no unique or exclusion constraint matching the ON CONFLICT specific |
| rb-738d19d4 | 738d19d4 | internal_error | internal_error | stmt 6: XX000 unsupported join alias expression |
| rb-6b858a36 | 6b858a36 | assertion | assertion | subselect.c: !IsA(node, SubPlan) |
| rb-8b29b8da | 8b29b8da | internal_error | internal_error | stmt 10: XX000 could not find pathkey item to sort |
| rb-d764d61a | d764d61a | wrong_result | wrong_result | stmt 7: 1 rows on fixed, 1 on buggy |
| rb-dfcf6fa7 | dfcf6fa7 | internal_error | internal_error | stmt 6: XX000 could not find pathkey item to sort |
| rb-a464d31f | a464d31f | assertion | assertion | relnode.c: join_clause_is_movable_into(rinfo, baserel->relids, joinrelids) |
| rb-852fd5b8 | 852fd5b8 | internal_error | internal_error | stmt 5: XX000 no relation entry for relid 3 |
| rb-a8bb1a77 | a8bb1a77 | assertion | assertion | rewriteManip.c: context->new_index != INVALID_VAR |
| rb-9d0adee8 | 9d0adee8 | none | none |  |
| rb-895c1b5f | 895c1b5f | none | none |  |
| rb-35d3f206 | 35d3f206 | internal_error | internal_error | stmt 21: XX000 variable not found in subplan target list |
| rb-1bf74c80 | 1bf74c80 | wrong_result | wrong_result | stmt 4: 4 rows on fixed, 3 on buggy |
| rb-7774249d | 7774249d | wrong_result | wrong_result | stmt 1: 1 rows on fixed, 1 on buggy |
| rb-0d995793 | 0d995793 | wrong_result | wrong_result | stmt 4: 0 rows on fixed, 1 on buggy |
| rb-d6eac691 | d6eac691 | wrong_result | wrong_result | stmt 0: 0 rows on fixed, 2 on buggy |
| rb-3b120b1e | 3b120b1e | missing_error | missing_error | stmt 0: fixed raises 22003 integer out of range |
note: rb-94e1385e raises errors on the fixed server: stmt 1: 2201E cannot take logarithm using base one; stmt 2: 2201E cannot take logarithm using base one
note: rb-41180127 raises errors on the fixed server: stmt 0: 22030 duplicate JSON object key value
note: rb-6b858a36 raises errors on the fixed server: stmt 3: 21000 more than one row returned by a subquery used as an expression
note: rb-3b120b1e raises errors on the fixed server: stmt 0: 22003 integer out of range; stmt 1: 22003 integer out of range; stmt 2: 22003 integer out of range

20/23 bugs observable; copy the symptoms into /home/bhuvi/cs5187-term-project/src/bench/reinjected/bugs.yaml
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$
```

bench verify: fixed / solo / injected servers

The benchmark

- Fixes of 31 Jul to 16 Oct 2026, each with a regression test
- 20 of 23 observable; only 9 visible to a crash-only fuzzer

Arm	Bugs found (of 20)
Random	0
Grammar	1
Grammar + oracles	**2**
+ LLM	**2**

Honest reading

- No arm found a wrong-result benchmark bug in 1 h
- The oracle advantage is shown by RQ2 (real wrong results found), not by RQ3

RESULTS

RQ4: the LLM extension

```
gow@Gowtham:~/cs5187-term-project/runs/memberB-rq1-llm$ tail -40 logs/B2_extend.txt
| finished | 9 |
| accepted | 4 |
| acceptance_rate | 0.444 |
| accepted_fragments | 4 |
| accepted_by_round | {"1": 2, "2": 1, "3": 1} |
| rounds_total | 24 |
| rejections_by_stage | {"validity": 15, "parse": 2, "coverage/seeds": 1, "reachability": 2} |
| tokens | {"input_tokens": 312197, "output_tokens": 156417, "cache_read_input_tokens": 46568, "cache_creation_input_tokens": 0} |
| cost_usd | 0.0 |
| mean_validity | 0.376 |
| coverage_gain_lines | 34 |
| gate_bug_candidates | 0 |
| rewrites_proposed | 7 |
| rewrites_accepted | 7 |
| rewrites_with_mismatch | 0 |
18:30:10 INFO d4c15309e7 round 1: rejected
  d4c15309e7-r2.json: gemini-3-flash-preview, 15659 in / 14514 out, 73s

=== extend invocation 15 (18:31:22) ===
1 commit(s) finished, 0 accepted; ledger: ../results/llm/ledger.jsonl
| metric | value |
|---|---|
| commits | 10 |
| finished | 10 |
| accepted | 4 |
| acceptance_rate | 0.4 |
| accepted_fragments | 4 |
| accepted_by_round | {"1": 2, "2": 1, "3": 1} |
| rounds_total | 25 |
| rejections_by_stage | {"validity": 16, "parse": 2, "coverage/seeds": 1, "reachability": 2} |
| tokens | {"input_tokens": 327856, "output_tokens": 170931, "cache_read_input_tokens": 46568, "cache_creation_input_tokens": 0} |
| cost_usd | 0.0 |
| mean_validity | 0.365 |
| coverage_gain_lines | 34 |
| gate_bug_candidates | 0 |
| rewrites_proposed | 7 |
| rewrites_accepted | 7 |
| rewrites_with_mismatch | 0 |
18:32:01 INFO d4c15309e7 round 2: rejected
=== extend finished: no prompt left ===
```

commit-directed extension run

Every prompt and response is stored, so the run can be replayed without the model.

4 / 10

commits accepted by the validation gate

+5

SQL features (p = 0.004), no coverage gain

What happened

- Gate rejected 16 of 25 rounds on server validity
- Fragments-only arm found upstream bug #19803
- One accepted rewrite changed the collation and caused a false positive; the gate now rejects explicit COLLATE
- Model: Gemini Flash (free tier); 10 of 25 mined commits used because of the daily quota

Evidence wall: runs on the team's machines

```
Successfully installed contourpy-1.4.0 cycler-0.12.1 fonttools-4.67.0 iniconfig-2.3.1 kiwisolver-1.5.1 matplotlib-3.11.2 num
py-2.5.4 packaging-26.3 pillow-13.0.0 pluggy-1.7.0 psycopg-3.3.6 psycopg-binary-3.3.6 pytest-9.1.1 python-dateutil-3.0.0 sci
py-1.18.1 tzdata-2026.5
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ export OUT=~/cs5187-term-project/runs/memberA-rq3
python3 -m pytest -q 2>&1 | tee $OUT/logs/01_pytest.txt
.....................................................................  [ 52%]
.....................................................................  [100%]
138 passed in 55.06s
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ tail -1 $OUT/logs/01_pytest.txt
138 passed in 55.06s
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$
```

unit tests pass

```
gow@Gowtham:~/cs5187-term-project/runs/memberB-rq1-llm$ cat logs/B1_selftest.txt
PASS  ok           plain query succeeds
PASS  syntax       parser rejections are SQLSTATE 42601
PASS  semantic     undefined table is 42P01
PASS  runtime      division by zero is 22012
PASS  continue     an error must not abort later statements of the same test case
PASS  timeout      statement_timeout must be set below 60 s
PASS  isolation-a  create a table ...
PASS  isolation-b  ... which must be gone in the next test case (no state leak)
PASS  terminate    orderly termination is not a crash; later statements are skipped
PASS  recover      executor reconnects after losing the connection

10/10 checks passed
```

executor self-test

```
10/10 checks passed
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$ python3 -m pgfuzz sample configs/rq3.yaml --name grammar_mm -n 3
-- testcase 0 (fresh)
CREATE TABLE n1 (c0 int GENERATED ALWAYS AS IDENTITY, c1 text[]);
INSERT INTO n1 (c1) SELECT string_to_array((x % 100)::text, '') FROM generate_series(1, 5) AS g(x);
ANALYZE n1;
MERGE INTO t3 AS a1 USING (SELECT 'ß'::text AS c0, CASE (-9223372036854775808)::bigint WHEN MIN ('15'::bigint) THEN CASE 1.5::numeric WHEN 2.25::numeric THEN int4ra
nge(2, 40) WHEN 42.5::numeric THEN '(,5]'::int4range ELSE 'empty'::int4range END WHEN (500)::bigint THEN '[3,)'::int4range ELSE '(,)'::int4range END AS c1 FROM (t3
AS a2 JOIN tp AS a3 ON ((a2.c0 = a3.c0) AND a2.c1)) WHERE a2.c1 GROUP BY a2.c4 HAVING BOOL_AND (a2.c1) LIMIT 1) AS a4 ON a1.c1 WHEN NOT MATCHED BY SOURCE THEN DELET
E;
SELECT '2024-12-31'::date AS c0, a1.c0 AS c1 FROM t0 AS a1, ROWS FROM (GENERATE_SERIES (1, 3), UNNEST ('{}'::text[])) AS a2 (c0, c1) WHERE TRUE ORDER BY 2 DESC;
-- testcase 1 (fresh)
UPDATE t1 AS a1 SET c0 = COALESCE (a1.c0, 0) WHERE NULLIF (COALESCE ((LEAST ((SELECT JSONB_AGG (((a2.c1 BETWEEN '25'::int AND '45'::int) AND TRUE) ORDER BY ((a2.c1
BETWEEN '25'::int AND '45'::int) AND TRUE)) FROM t1 AS a2 WHERE ('NaN'::text IS NULL)), JSONB_BUILD_OBJECT ('k', LEAST ('{1,2,3,4}'::int[], CASE WHEN TRUE THEN '{}'
::int[] ELSE ('{1,2,3}'::int[] || COALESCE (CASE ARRAY[1, 2, 3] WHEN '{1,2,3}'::int[] THEN a1.c0 WHEN '{}'::int[] THEN a1.c0 ELSE a1.c0 END, 0)) END))) IS NULL), (R
OW ((GREATEST ('{"k": 0, "a": [3, 4, 5]}'::jsonb, '{"k": 1, "a": [1, 2]}'::jsonb) -> 'k'), LEAST ('1 day'::interval, COALESCE ('-1 month'::interval, '7 days'::inter
val))) <> ROW (CASE WHEN FALSE THEN '{"k": -1, "a": [1, 2, 3]}'::jsonb ELSE (SELECT JSON_AGG ('2000-01-01'::date ORDER BY '2000-01-01'::date DESC)::jsonb FROM t4 AS
 a3) END, (CASE WHEN GREATEST (TRUE, TRUE) THEN '00:00:00.000001'::interval ELSE '1 day'::interval END * 0)))), ((REVERSE ('a'::text) ^@ a1.c1) OR COALESCE (((a1.c1
 COLLATE "C") >= GREATEST ('abc'::text, a1.c1)), NULLIF (FALSE, ('{"k": -1, "a": [1, 2, 3]}'::jsonb ? 'x')))));
SELECT FALSE AS c0, CARDINALITY (MIN (REGEXP_SPLIT_TO_ARRAY (a1.c1, '[ab]'))) AS c1, (- 5e-324::float8) AS c2, '00:00:00.000001'::interval AS c3 FROM JSON_TABLE (TO
_JSONB (('-1 month'::interval * 1)), '$[*]' COLUMNS (c0 int PATH '$.k', c1 text PATH '$')) AS a1 GROUP BY a1.c0 ORDER BY 4 NULLS FIRST;
-- testcase 2 (fresh)
SELECT COALESCE (SUBSTR (LOWER (CASE LEAST ('2000-01-01 00:00:00+00'::timestamptz, '2024-01-01 15:00:00+00'::timestamptz) WHEN '1999-12-31 23:59:59.999999+05:30'::t
imestamptz THEN '[1,10)'::text WHEN '2024-01-01 05:00:00+00'::timestamptz THEN JSON_VALUE (CASE WHEN (NOT TRUE) THEN JSON_QUERY (CASE 'x%'::text WHEN FORMAT ('%s|%s
', '(,)'::int4range, (-1)::bigint) THEN CASE '2000-01-01 00:00:00+00'::timestamptz WHEN '1999-12-31 23:59:59.999999+05:30'::timestamptz THEN '{"k": 1, "a": [1, 2]}'
::jsonb WHEN 'epoch'::timestamptz THEN '{"k": 0, "a": [3, 4, 5]}'::jsonb ELSE '{"k": 2, "a": [0]}'::jsonb END WHEN '23x'::text THEN '{"k": 1, "a": [1, 2]}'::jsonb E
LSE '{"k": 2, "a": [0]}'::jsonb END, '$.a[last]') ELSE '{"k": 0, "a": [3, 4, 5]}'::jsonb END, '$.a.size()') ELSE 'A'::text END), NTILE (3) OVER (PARTITION BY JSON_A
RRAY ('6.3333333333333333'::float8, '50'::float8 RETURNING jsonb), 1::bigint ORDER BY '350:00:00'::interval, '{"k": 0, "a": [3, 4, 5]}'::jsonb NULLS LAST)), (ABS ('1
2.00'::numeric))::text) AS c0, '2.5'::numeric AS c1 FROM (SELECT FALSE AS c0, a1.c1 AS c1, '2024-01-26'::date AS c2 FROM ti AS a1 GROUP BY GROUPING SETS ((a1.c1), (
))) AS a2 WHERE ('[3,)'::int4range -|- INT4RANGE (LEAST (COALESCE (('25'::int + CARDINALITY (LEAST (ARRAY_REMOVE ('{1,2,3}'::int[], 500), '{}'::int[]))), LENGTH ('A
'::text)), 0), GREATEST ((((SELECT MAX (a3.c0) FROM ti AS a3 WHERE ('%'::text BETWEEN SYMMETRIC a3.c1 AND SUBSTR (REPLACE ('x'::text, a3.c1, a2.c1), '4'::int))) % 1
00) * (GREATEST ((-15), (NULLIF ('50'::int, 60) + 38)) % 100)), 0))) HAVING ('empty'::int4range >> RANGE_INTERSECT_AGG ('[3,)'::int4range));
DELETE FROM t3 AS a1 USING t0 AS a2 WHERE (EXISTS (SELECT a1.c1 AS c0, '15'::bigint AS c1, COUNT (*) AS c2 FROM v0 AS a3 WHERE a1.c1 GROUP BY a3.n HAVING (TRUE OR a
1.c1))) RETURNING a2.c0 AS c0, CASE WHEN CASE INT4RANGE (LEAST (('2024-02-29'::date - a1.c2), 0), GREATEST (GREATEST (a2.c0, 7), 0)) WHEN CASE WHEN a1.c1 THEN INT4R
ANGE (LEAST (a2.c0, 10), GREATEST (a2.c0, 10), '[]') ELSE int4range(2, 40) END THEN a1.c1 WHEN '(,5]'::int4range THEN (NULL::jsonb @> TO_JSONB (ARRAY['1'::text])) E
LSE (COALESCE ('[1,10)'::int4range, '[0,0]'::int4range) &> INT4RANGE (LEAST (ASCII (SPLIT_PART (a2.c1, ',', (-1))), 10), GREATEST (a1.c0, 10), '(]')) END THEN '-0':
:float8 ELSE (100.0::float8 - '6.333333333333333'::float8) END AS c1, '12.00'::numeric AS c2;
bhuvi@BlazingBhuvi:~/cs5187-term-project/src$
```

grammar sample: generated SQL, validity

```
bhuvi@BlazingBhuvi:~/cs5187-term-project$ cat ~/cs5187-term-project/runs/memberA-rq3/logs/A5_attribute.txt
| bug | symptom | random | grammar | grammar_mm | grammar_llm |
|---|---|---|---|---|---|
| rb-11b56ee2 | assertion | 0/5 | 0/5 | 1/5 (2901s) | 0/5 |
| rb-94e1385e | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-41180127 | missing_error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-b28a2b2b | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-d8444163 | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-62e46c00 | error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-738d19d4 | internal_error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-6b858a36 | assertion | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-8b29b8da | internal_error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-d764d61a | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-dfcf6fa7 | internal_error | 0/5 | 5/5 (458s) | 4/5 (509s) | 0/5 |
| rb-a464d31f | assertion | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-852fd5b8 | internal_error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-a8bb1a77 | assertion | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-35d3f206 | internal_error | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-1bf74c80 | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-7774249d | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-0d995793 | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-d6eac691 | wrong_result | 0/5 | 0/5 | 0/5 | 0/5 |
| rb-3b120b1e | missing_error | 0/5 | 0/5 | 0/5 | 0/5 |
| **bugs found** | | **0/20** | **1/20** | **2/20** | **0/20** |

native failures (reproduce on the fixed server): 16; unattributed: 20
bhuvi@BlazingBhuvi:~/cs5187-term-project$
```

RQ3 attribution by replay on solo servers

EXECUTION

What went wrong, and how we handled it

Incidents

- One worker's server failed crash recovery after an assertion: 14 trials re-run with the same seeds
- Laptop sleep during a campaign: trial clocks are monotonic, budget unchanged

Substitutions

- LLM arm ran on Gemini Flash (free tier) via the tool's manual mode, no source change
- Daily quota: 10 commits instead of 30

False positives

- Four new oracle false-positive classes appeared in the campaigns
- Fixed afterwards and tested on stored reproducers; not re-measured

Reported honestly

- Raw precision 0.16 is logged as measured
- RQ3 shows no oracle advantage; we say so

PLAN

Next steps to 13 November

When	What
30 Oct to 5 Nov	Verify the 6 open candidates (5 RLS, 1 corr); follow up upstream threads; re-run BUG-01/02 reproducers
2 to 6 Nov	Meeting 3: full result tables, paper draft
12 Nov	Presentation
13 Nov	Paper and repository due

Questions for you

- Do committer-confirmed bugs (one fixed) meet the top band, even though the benchmark shows no oracle advantage?

Thank you. Questions?

github.com/v28k/cs5187-term-project